Name		Status		Logistics								Software Requirements	Description
ID	Task	Subtask	Status	Blocking Point	Priority	Discipline	Specific Discipline	Assigned	Time Estimate	Time Taken	Update Date	Development Package	Description
10	Track Design	4	Complete		P0	Preparation	Track	Ferréol	30	70	2024-11-09	NA	Make initial 5m X 5m grid for the track
10.1	Track Design	Material Research	Complete		P0	Preparation	Track	Ferréol	6	8	2024-10-20	NA	Find the appropriate material for the track, for the lines. Look up prices, compare feasibility and quality of different options
10.2	Track Design	Material Purchase	Complete		P0	Preparation	Track	Ferréol	8	8	2024-10-26	NA	Visit stores for quotes, purchase material and tools required, bring to McGill
10.3	Track Design	Esquisse	Complete		P0	Preparation	Track	Ferréol	8	14	2024-11-02	NA	Draw general outline for the track, take measurements and place markings precisely before permanent
10.4	Track Design	Réalisation	Complete		P0	Preparation	Track	Ferréol	8	40	2024-11-17	NA	Apply permanent markings to the track
11	IMU Data - Bosch Sensor	5	Complete		P1	Hardware	Electronics	Xin Ya	54	30	2024-11-22	Sensing	Obtain accurate data from the IMU
11.1	IMU Data - Bosch Sensor	Understand Sensor	Complete		P1	Hardware	Electronics	Xin Ya	4	4	2024-10-26	Sensing	Read up on the documentation and understand how the sensor works. Be able to explain to anyone on the team
11.2	IMU Data - Bosch Sensor	Read Values with STM32	Complete		P1	Hardware	Electronics	Xin Ya	10	10	2024-11-02	Sensing	Use STM32 to read values from the sensor and print them. Must be able to parse and get all data output by sensor
11.3	IMU Data - Bosch Sensor	Test Quality of Values	Complete		P1	Hardware	Electronics	Xin Ya	20	10	2024-11-09	Sensing	Test the quality of the data output and its limits. Prepare testing document and data analysis
11.4	IMU Data - Bosch Sensor	UART with Jetson	Complete		P1	Hardware	Electronics	Xin Ya	10	4	2024-11-23	Sensing	Establish serial communication with the Jetson. Test for different use cases and make sure to be able read clearly
11.5	IMU Data - Bosch Sensor	Script/Flash	Complete		P1	Hardware	Electronics	Xin Ya	10	2	2024-11-30	Sensing	Write script that automatically reads IMU data and publishes it over UART. Must be robust with reset.
12	Motor Control - Input Acceleration	7	Not required		P2	Hardware	Electronics	Xin Ya	45		2025-01-11	Sensing	Develop a new and better way to control the output to the motor
12.1	Motor Control - Input Acceleration	Understand current control	Complete		P2	Hardware	Electronics	Xin Ya	4	8	2024-12-26	Sensing	Find out whatever is going on with the current script. Why speed? How is it converted? What does the motor get? Why did they do it like that? Research how the motor works and its specs.
12.2	Motor Control - Input Acceleration	Work out our needs	Complete		P2	Hardware	Electronics	Xin Ya	1	4	2025-01-04	Sensing	Discuss with software and hardware to see what is best for us. What is used in industry? What gives us the best response? What is feasible and not based on previous task.
12.3	Motor Control - Input Acceleration	Develop Script	Complete		P2	Hardware	Electronics	Xin Ya	8		2025-01-11	Sensing	Write out a script that implements what has been decided. Provide adequate documentation
12.4	Motor Control - Input Acceleration	Test Script	Not required		PN	Hardware	Electronics	Xin Ya	8		2025-01-11	Sensing	Test out the script with the motor. See the response and provide testing documentation. Iterative approach to work out algorithm
12.5	Motor Control - Input Acceleration	Combine with IMU script & Flash	Not required		PN	Hardware	Electronics	Xin Ya	8		2025-01-18	Sensing	Combine the script with the IMU script to be able to run both at once.
12.6	Motor Control - Input Acceleration	Test combined script	Not required		PN	Hardware	Electronics	Xin Ya	8		2025-01-11	Sensing	Test out combined script. Make sure no errors, lost data, stalling, delays, corruption
12.7	Motor Control - Input Acceleration	UART with Jetson	Not required		PN	Hardware	Electronics	Xin Ya	8		2025-01-11	Sensing	Test out combined script. Make sure no errors in communication, crashing or lost signals
13	Technical Challenge Path	4	In Progress		P4	Software	Path Planning	Yu Qing	38		2025-02-22	Planning	Optimize path for the technical challenge
13.1	Technical Challenge Path	Survey of available approaches	Complete		P4	Software	Path Planning	Yu Qing	8		2024-11-09	Planning	What are the available options? Pros and Cons of each? Specificities to our competition? Be able to present and explain them to the team to discuss. Select 2 or 3 approaches to attempt
13.2	Technical Challenge Path	Develop algorithm following approach	In Progress		P4	Software	Path Planning	Yu Qing	12		2024-12-07	Planning	Develop algorithm in python, visualisation required. Summary of results, problems and what they imply. Take into account possible placement, trajectory, re-adjustment, etc.
13.3	Technical Challenge Path	Test algorithm (Graph)	Waiting	Dependent on prior task	P4	Software	Path Planning	Yu Qing	12		2025-01-11	Planning	Iterative testing, in combination with previous task/algorithm development
13.4	Technical Challenge Path	Simulator testing of paths	Waiting	Dependent on prior task	P4	Software	Path Planning	Yu Qing				Planning	Run the car in the simulator such that it follows all the possible paths from all starting locations. Identify any weaknesses of issues of the following types : time, obstacles, speed, respect of road regulation.
13.5	Technical Challenge Path	Integrate algorithm	Waiting	Dependent on prior task	P4	Software	Path Planning	Yu Qing	6		2025-02-08	Planning	Integrate the algorithm with the other packages. Fix any merging issues and situations in which to apply it.
14	RealSense - Launch	3	Complete		P2	Hardware	Devices	Malo & Simon	36		2024-11-01	Sensing	Fix realsense launch issues, find method to launch reliably and consistently
14.2	RealSense - Launch	Understand provided code	Complete		P2	Hardware	Devices	Malo	8	8	2024-10-27	Sensing	How does the current script launch the realsense and send the data? ROS? Serial communication? Explanation for everyone
14.3	RealSense - Launch	Survey of problems	Complete		P2	Hardware	Devices	Malo	12	12	2024-11-02	Sensing	Discuss issues that aros last year. Detail them and find out what causes them. Documentation and explanation for issues to explain to team mates.
14.4	RealSense - Launch	Find method to launch realsense	Complete		P2	Hardware	Devices	Simon	16	20	2024-11-09	Sensing	Find the method that enables the realsense launch with reliability. New script? Settings within current script? Required manipulations? Document procedure.
15	IMU Data - RealSense	4	Complete		P3	Hardware	Devices	Malo & Xin Ya	30		2024-12-16	Sensing	Obtain accurate data from the RealSense IMU
15.1	IMU Data - RealSense	Understand IMU function	Complete		P3	Hardware	Devices	Malo	4	4	2024-11-09	Sensing	Understand the way the RealSense IMU works. Specificities, ways it publishes data and how it communicates
15.2	IMU Data - RealSense	Identify issues with IMU data	Complete		P3	Hardware	Devices	Xin Ya	10	10	2024-11-23	Sensing	Identify the issues with obtaining the data. Survey data from previous year, test situations and launching
15.3	IMU Data - RealSense	Find method to acquire IMU data	Complete		P3	Hardware	Devices	Xin Ya	10	4	2024-11-30	Sensing	Find the method that enables us to get IMU data from the RealSense reliably. New script? Settings within current script? Required manipulations? Document procedure.
15.4	IMU Data - RealSense	Test Quality of Values	Complete	Filtering to get the YAW yields unstable values	P1	Hardware	Devices	Xin Ya	6		2024-11-30	Sensing	Test the quality of the data output and its limits. Prepare testing document and data analysis
16	Documentation - Hardware	2	In Progress		P2	Hardware	Documentation	Malo	8		2025-01-01	NA	Document this year's hardware
16.1	Documentation - Hardware	Interaction Diagram	In Progress		P2	Hardware	Documentation	Malo	4		2024-12-16	NA	Create architecture for hardware. Detail with all components and connections.
16.2	Documentation - Hardware	Organize previous year documents	Complete		P3	Hardware	Documentation	Malo	4	4	2024-11-16	NA	Organize last year's documents for further reference and help
17	Documentation - Software	2	In Progress		P2	Software	Documentation	Simon	22		2025-01-01	Multiple	Document this year's software
17.1	Documentation - Software	Re-Organize GitHub RePo	Complete		P2	Software	Documentation	Simon	10		2024-11-30	Multiple	Organize the GitHub Repo such as to make it easier to navigate and develop. Needs to be done before sharing with BFMC
17.2	Documentation - Software	Write ReadME for all	In Progress		P4	Software	Documentation	Simon	12		2025-01-01	Multiple	Write ReadMe for all repositories, to make it understandable and portable. All important information should be contained such that a newcomer knows how to install and use
18	Integration Testing	3	In Progress		P1	Hardware	Testing	Malo	16		2025-01-01	Multiple	Ongoing integration testing for all algorithms
18.1	Integration Testing	Previous Year Running	Complete		P1	Hardware	Testing	Malo	6		2024-11-16	Multiple	Get last year's algorithms to work
18.2	Integration Testing	New Car Kit/Running	Complete		P1	Hardware	Testing	Malo	10		2024-12-07	Multiple	Get last year's algorithms to work on the new Bosch provided Harware
18.3	Integration Testing	General Testing/Bug Fixing	Complete		P1	Hardware	Testing	Malo/Simon	4		2024-11-13	Multiple	General testing to find issues with code, hardware. Test for accuracy, reliability,etc.
19	Steering Calibration	2	Complete		P2	Hardware	Testing	Malo/Ferréol	21		2024-12-16	Sensing	Calibrate the steering for accuracy
19.1	Steering Calibration	Test the actual steering accuracy	Complete		P0	Hardware	Testing	Malo/Ferréol	3	8	2024-11-23	Sensing	Test the current quality of the steering
19.2	Steering Calibration	Calibrate the steering	Complete		P0	Hardware	Testing	Malo/Ferréol	10	40	2024-11-23	Sensing	Calibrate and ajust the sensitivity of the steering
19.3	Steering Calibration	Characterize the motor and create speed curve	Complete		P0	Hardware	Testing	Malo/Ferréol	8		2024-12-31	Sensing	Measure multiple different turning radii from given range of input PWM values, fit curve to the point cloud to get relationship between the turning radius and input PWM
19.4	Steering Calibration	Write function to compute speed from curve	Complete		P0	Hardware	Testing	Malo	8		2024-12-31	Sensing	Modify the code on the STM32 to be able to compute the PWM given an input Steering angle from the equations found in the preceding task
20	Chassis Design - Development	3	Complete		P2	Hardware	Chassis	Ferréol	14		2024-11-09	NA	Design a chassis that makes the installing and removing of all boards (jetson, STM32) easy and efficient. Must take into account cable management, rigidity, stability.
20.1	Chassis Design - Development	Measurements and survey	Complete		P2	Hardware	Chassis	Ferréol	2		2024-10-20	NA	Survey of car dimensions, installation and anchor points, dimensions of boards and of chasis.
20.2	Chassis Design - Development	Design hypothesis	Complete		P2	Hardware	Chassis	Ferréol	2		2024-10-26	NA	Rough ideas and brainstorming, different approaches and possible mechanisms that could work
20.3	Chassis Design - Development	Modelling	Complete		P2	Hardware	Chassis	Ferréol	6		2024-10-26	NA	Modelling and design of actual parts
20.4	Chassis Design - Development	Printing and testing	Complete		P2	Hardware	Chassis	Ferréol	4		2024-11-02	NA	Printing and testing of parts to see if functional design
21	Communication with Car	4	Complete		P2	Software	Running	Simon	56		2024-02-01		Implement a way to communicate with the car that minimizes delays and gives us real-time feedback
21.1	Communication with Car	Identify source of current problems	Complete		P2	Software	Running	Simon	6		2024-12-07	Multiple	Identify the reasons for which communication is very slow right now
21.2	Communication with Car	Research alteratives & report	Complete		P2	Software	Running	Yu Qing	10		2025-01-05	Multiple	Research alternative methods of communication with the jetson, that would enable real-time streaming of data
21.3	Communication with Car	Implement test scripts	Complete		P2	Software	Running	Yu Qing	20		2025-01-19	Multiple	Implement test scripts and measure the delays of communication between devices
21.4	Communication with Car	Adapt to the dashboard	Complete		P2	Software	Running	Yu Qing	20		2025-02-01	Multiple	Adapt the dashboard so that the source of the data can be from the stream of information and not through ROS
21.5	Communication with Car	Adapt to services	Complete		P2	Software	Running	Yu Qing	20		2025-02-01	Multiple	Adapt the services such that they are called through TCP and not through ROS
22	Lane Center Relocalization	2	Complete		P2	Software	Localization	Simon	30		2025-02-11	Planning	Adapt lane detection such that it uses the lanes to relocalize the car in the center of the lane
22.21	Lane Center Relocalization	Implement algorithm to estimate current position	Complete		P2	Software	Localization	Simon			2025-02-01	Planning	Implement an algorithm that can take the current lane center and estimate the car's position on the map with respect to this lane center
22.2	Lane Center Relocalization	Lane detection relocalization in Simulator	Complete		P2	Software	Localization	Simon	10		2025-02-01	Planning	Test the relocalization in the simulator to see if the relocalization is accurate
22.3	Lane Center Relocalization	Testing lane detection relocalization on track	Complete		P2	Software	Localization	Simon	20		2025-02-11	Planning	Test the relocalization on the real track to see how the relocalization affects the performance

Name		Status		Logistics								Software Requirements	Description
ID	Task	Subtask	Status	Blocking Point	Priority	Discipline	Specific Discipline	Assigned	Time Estimate	Time Taken	Update Date	Development Package	Description
22.4	Lane Center Relocalization	Adjust and improve lane detection from last year	In Progress		P2	Software	Localization	Yu Qing			2025-03-30	Planning	Look into the previous year's lane detection made by Malo, understand the code and make it work
22.5	Lane Center Relocalization	Fit the lane equations and extract waypoints	In Progress		P2	Software	Localization	Yu Qing			2025-03-30	Planning	Extract a correct polynomial from the lanes and waypoints for path planning
22.6	Lane Center Relocalization	Test in the simulator	Waiting	Dependent on prior task	P3	Software	Localization	Yu Qing			2025-03-30	Planning	Test the implementation of the lane detection inside the simulator
22.7	Lane Center Relocalization	Test on the real car	Waiting	Dependent on prior task	P4	Software	Localization	Yu Qing			2025-03-30	Planning	Test the implemenation of the lane detection on the real track
23	Speed Calibration	5	Complete		P2	Hardware	Running	Malo & Ferréol	10		2024-12-08	NA	Make sure that the commands we are sending to the car reflect the actual behaviour of the car
23.1	Speed Calibration	Test the current speed accuracy	Complete		P1	Hardware	Running	Malo & Ferréol	4		2024-11-29	NA	Test the calibration that Bosch made on the car. Check for error values and accuracy
23.2	Speed Calibration	Understand the code for controlling speed	Complete		P1	Hardware	Running	Malo & Ferréol	3		2024-12-21	NA	Understand the code so that adjustments are based on the internal workings of the embedded system platform
23.2	Speed Calibration	Tune the values for speed	Complete		P1	Hardware	Running	Malo & Ferréol	8		2024-12-31	NA	Tune the values based on the test results and the understood code from embedded platform
23.3	Speed Calibration	Characterize the motor and create speed curve	Complete		P1	Hardware	Running	Malo & Ferréol	8		2024-12-31	NA	Measure multiple different speeds from given range of input PWM values, fit curve to the point cloud to get relationship between the speed and input PWM
23.4	Speed Calibration	Write function to compute speed from curve	Complete		P1	Hardware	Running	Malo	8		2024-12-31	Embedded System	Modify the code on the STM32 to be able to compute the speed from the equations found in the preceding task
24	SLAM Realsense	5	In Progress		P2	Software	Localization	Simon	59.00		2025-01-11	Sensing	Implement simultaneous localization and mapping using the realsense camera
24.1	SLAM Realsense	Algorithm Research	In Progress		P2	Software	Localization	Simon	10.00		2024-12-26	Sensing	Look into the slam algorithm and if it is feasible for our car and for our current use case
24.2	SLAM Realsense	Testing different librairies	In Progress		P2	Software	Localization	Simon	10.00		2025-01-04	Sensing	Research different librairies that are currently available for the intel realsense camera that make use of both the IMU, the Depth camera and the wide angle camera
24.3	SLAM Realsense	Simulator Testing	In Progress		P2	Software	Localization	Simon	20.00		2025-02-01	Sensing	Do some testing on the simulator to see if the librairies can actually be used to localize the car and if the results are promising
24.4	SLAM Realsense	Jetson Runtime Test	Waiting	Segmentation fault when launching ROS-Realsense	P2	Software	Localization	Simon	4.00		2025-01-11	Sensing	Do some runtime tests on the jetson to see if it is able to work throught the algorithm at a steady pace and is compatible with the object recognition model
24.5	SLAM Realsense	Integration Testing	Waiting	Dependent on prior task	P2	Software	Localization	Simon	15.00		2025-01-18	Sensing	Do an integration test to see if it is possible to use this algorithm on the real track
25.1	Power Distribution Board Replacement	4	In Progress		P2	Hardware	Running	Xin Ya	6.00		2025-01-05	NA	Attempt to isolate which component is causing the absence of voltage at the terminals. Use both the gerber files on altium to look at the circuit diagram and probe points on the actual board
25.2	Power Distribution Board Replacement	Research alteratives, potential solutions & report	Complete		P2	Hardware	Running	Xin Ya	8.00		2025-01-12	NA	Survey the different options available to us and evaluate their feasibility. Rank them in terms of their worth to the project considering cost, ease of implementation, and time taken to implement. Present the report to the team
25.3	Power Distribution Board Replacement	Temporary replacement of old board	Complete		P1	Hardware	Running	Xin Ya			2025-01-26	NA	Remove the motor wires from the new board and connect them to the old board so that we can use only the old board in the testing of the car, and therefore only one battery
25.4	Power Distribution Board Replacement	Design, submissions, purchase or ordering of solutions.	Complete		P2	Hardware	Running	Xin Ya	4.00		2025-01-26	NA	Send out any submissions and contact suppliers if needed, order components
25.5	Power Distribution Board Replacement	Installation of new powerboard	Waiting	Added task 25.6, dependent on this prior task	P2	Hardware	Running	Xin Ya	10.00		2025-02-01	NA	Test out the chosen solution and subsequently install it on the car once everything has been verified to be working
25.6	Power Distribution Board Replacement	Re-design replacement with included voltage regulation	In Progress		P1	Hardware	Running	Xin Ya			2025-03-23	NA	Motor turning speed depends on the input voltage, therefore need correct voltage regulation to ensure that the calibration is accurate
26	Track Design - Qualification	3	In Progress		P2	Hardware	Track	Ferréol	34.00		2025-02-09	NA	Add the required parts of the track ahead of the qualification run
26.1	Track Design - Qualification	Design - Calculate required track material and separation	Complete		P2	Hardware	Track	Ferréol	4.00		2025-01-26	NA	Separate the material, give estimate of quantity required for purchase
26.2	Track Design - Qualification	Esquisse	Complete		P2	Hardware	Track	Malo & Ferréol	10.00		2025-02-02	NA	Draw general outline for the track, take measurements and place markings precisely before permanent
26.3	Track Design - Qualification	Réalisation	In Progress		P2	Hardware	Track	Malo & Ferréol	20.00		2025-02-09	NA	Apply permanent markings to the track
27	Model Based Steering	4	In Progress		P1	Hardware	Path Planning	Simon			2025-02-09	Embedded System	Implement computation of the steering angle based on the desired yaw. This desired yaw is computed by the use of the internal kinematic bicycle model and outputs from the MPC controller.
27.1	Model Based Steering	Understand compuation of MPC	Complete		P1	Hardware	Path Planning	Malo			2025-01-12	Embedded System	Understand how the MPC controller supplies information that is used by the kinematic bicycle model, how the yaw is computed and what are the outputs of the RK4 computation
27.2	Model Based Steering	Implement steering to compute angle from expected yaw	Complete		P1	Hardware	Path Planning	Malo			2025-01-26	Embedded System	Implement a functionnality that enables the desired yaw to be sent over serial and used as an input to a function on the STM32. This function uses a PID to match the current yaw to desired yaw
27.3	Model Based Steering	Extensive testing to tune the PID and evaluate quality	In Progress		P1	Hardware	Path Planning	Malo			2025-03-04	Embedded System	Test out and tune the PID to get optimal following of the desired yaw without instability.
27.4	Model Based Steering	Implement Extended Bicycle Model	In Progress		P2	Hardware	Path Planning	Malo			2025-03-04	Embedded System	Replace the Kinematic bicycle model with the extended bicycle model which accounts for the vehicle's center of mass with respect to front and rear axes
27.5	Model Based Steering	Document progress and functionality	Waiting	Dependent on prior task	P1	Hardware	Path Planning	Malo			2025-03-16	Embedded System	Document the functionnality of the algorithm and evaluate the quality of the performance.
28	Jetson SSD replacement	3	Complete		P3	Hardware	Devices	Yu Qing			2025-02-01	Multiple	Install an SSD in place of the SD card for faster building times, responses and less computational power used up
28.1	Purchase appropriate SSD	Purchase an SSD that is compatible with the Jetson	Complete		P3	Hardware	Devices	Yu Qing			2025-01-26	Multiple	Look into which SSDs are compatible with our model of the Jetson, and purchase the appropriate one
28.2	Clone SD card and install Jetpack	Install all dependencies for appropriate function	Complete		P3	Hardware	Devices	Yu Qing			2025-01-26	Multiple	Clone the SD, install the appropriate Jetpack version on the SSD and make sure all required librairies and dependencies are adequately accounted for
28.3	Test all functionalities of the Jetson	Test out the Jetson with all software	Complete		P3	Hardware	Devices	Yu Qing			2025-02-01	Multiple	Test out all the code and functionalities of the Jetson to make sure everything functions appropriately and that there are no missing dependencies
29	Object Detection Enhancement	4	In Progress	Low Priority	P2	Software	Path Planning	Xin Ya			2025-03-08	Sensing	Re-train the model detection algorithm to account for the variety of scenarios in which the current model has weaknesses and to over a greater range of scenarios
29.1	Object Detection Enhancement	Identify and Summarize weaknesses	Complete	Low Priority	P2	Software	Path Planning	Xin Ya			2025-02-01	Sensing	Identify the current weaknesses of the model : i.e. which signs does it have more trouble with? At what distance is it weak from? Are there specific angles that inhibit it from detecting something?
29.2	Object Detection Enhancement	Generate or Create Data Set for Training	Complete	Dependent on prior task	P2	Software	Path Planning	Xin Ya			2025-02-16	Sensing	Create the appropriate data set for training the model to account for its current failures
29.3	Object Detection Enhancement	Formalize Test Set for Validation	Complete	Dependent on prior task	P2	Software	Path Planning	Xin Ya			2025-02-17	Sensing	Create a formal test set that contains a standardized set of images, lighting conditions, environments, positions and signs to validate and qualify the performance of the model
29.4	Object Detection Enhancement	Train the Data Set	In Progress	Dependent on prior task	P2	Software	Path Planning	Xin Ya			2025-03-01	Sensing	Train the model using dataset augmentation.
29.5	Object Detection Enhancement	Export the model to the Jetson	Waiting	Dependent on prior task	P2	Software	Path Planning	Xin Ya			2025-03-08	Sensing	Put the model on the jetson and evaluate it's detection accuracy
29.6	Object Detection Enhancement	Annotate the Test Set for MAP calculation	In Progress		P3	Software	Path Planning	Xin Ya			2025-04-06	Sensing	Upload the current test set to the drive so that Simon and Yu Qing can help with the annotation
29.7	Object Detection Enhancement	Create document with training specifications	In Progress		P3	Software	Path Planning	Xin Ya			2025-04-06	Sensing	Specify the diferent ratios of images within the training set. Which type of object, how many of them are there, what image augmentation techniques are used, etc.
29.8	Object Detection Enhancement	Create framework for training and testing	Waiting	Dependent on prior task	P3	Software	Path Planning	Xin Ya			2025-04-06	Sensing	Provide all the steps to training the model accurately and to produce the best results.
30	Headlight installation	4	Waiting	Low Priority	P4	Hardware	Electronics	Xin Ya			2025-03-08	Embedded System	Install headlights on the car to increase the accuracy of object detection, as lower brightness may be a cause of failuer to detect objects in front of it
30.1	Headlight installation	Document and Evaluate current headlights	Waiting	Dependent on prior task	P4	Hardware	Electronics	Xin Ya			2025-02-23	Embedded System	Evaluate and document the current leftover headlights from the previous year. What is their specified voltage and current draw? How are the activated? How much control do we have over them?
30.2	Headlight installation	Evaluate feasibility of installation & requirements	Waiting	Dependent on prior task	P4	Hardware	Electronics	Xin Ya			2025-02-23	Embedded System	Design an installation to install the headlights on the car given our current hardware and constraints
30.2	Headlight installation	Design of installation mounts	Waiting	Dependent on prior task	P4	Hardware	Chassis	Ferréol			2025-03-01	Embedded System	Design a mount to install the headlights in the appropriate place so that they illuminate the appropriate sections of the field of view
30.3	Headlight installation	Install the mounts and evaluate the added benefit	Waiting	Dependent on prior task	P4	Hardware	Electronics	Ferréol			2023-03-08	Embedded System	Install the headlights on the car and evaluate how well they work in combination with the other components.
31	Sign Detection Relocalization	4	In Progress		P1	Software	Localization	Simon			2025-02-15	Planning	Adjust and tune the sign detection relocalization algortihm such that the relocalization is as accurate as can be.
31.1	Sign Based Relocalization	Simulator Testing : Find optimal height	Complete		P1	Software	Localization	Simon			2025-02-08	Planning	Through testing in the simulator, find the optimal position of the realsense with respect of the car. Identify the parameters required for the accurate computation of the distance from the object.
31.2	Sign Based Relocalization	Mount Design : Design and Print	Complete		P1	Hardware	Chassis	Ferréol			2025-02-08	Planning	Model and build a mount to install the camera at the height and angle required from the previous task
31.3	Sign Based Relocalization	Unit testing : Accuracy of solution	Complete		P1	Preparation	Testing	Malo & Simon			2025-02-08	Planning	Test the relocalization ability of the car. First proceed with unit tests to estimate the distance and error, the procede with real running tests to see if there is a qualitatively noticeable improvement in the performance of the
31.4	Sign Based Relocalization	Analysis of results	In Progress		P1	Preparation	Testing	Malo & Ferréol			2025-02-15	Planning	Minimize discrepancies, evaluate quantitatively and qualitatively the error between actual position and estimated position
32	Traffic Light Classifier	4	Complete		P3	Software	Localization	Simon			2025-03-01	Sensing	Implement an algorithm that can identify the color of the traffic light after the initial object detection has identified the trafffic light.
32.1	Traffic Light Classifier	Research into available algorithms	Complete		P3	Software	Localization	Simon			2025-02-09	Sensing	Research into developped methods to identify traffic lights, contour detection, optimal thresholding for the color and brightness detected
32.2	Traffic Light Classifier	Develop classifier	Complete		P3	Software	Localization	Simon			2025-02-15	Sensing	Implement and algorithm that uses a combination of the previously identified qualifiable algorithms for our use case
32.3	Traffic Light Classifier	Unit testing, iterative tuning of classifier	Complete		P3	Software	Localization	Simon			2025-02-15	Sensing	Test the quality of the detection using the RealSense camera feed and iteratively tune the classifier to get the best results
32.4	Traffic Light Classifier	Integration testing on car in action	Complete	Dependent on prior task	P3	Software	Localization	Simon			2025-03-01	Sensing	Test the classifier on the moving car during a test run.
33	Frenet-Space Coordinates	4	In Progress		P3	Software	Localization	Simon			2025-04-01	Planning	Transition from an absolute coordinate system to locate the position of the vehicle to the Frenet-Space coordinate system. This will increase the flexibility of the path planning by shifting the way points relatively to the vehicle rather than relative to the track
33.1	Frenet-Space Coordinates	Understand the Frenet-Space Coordinates	Complete		P3	Software	Localization	Simon			2025-02-16	Planning	Understand how the Frenet-Space is described and how changes within it are characterized

Name		Status		Logistics								Software Requirements	Description
ID	Task	Subtask	Status	Blocking Point	Priority	Discipline	Specific Discipline	Assigned	Time Estimate	Time Taken	Update Date	Development Package	Description
33.2	Frenet-Space Coordinates	Implement function to transition from absolute to Frenet-Space	In Progress		P3	Software	Localization	Simon			2025-03-01	Planning	Implement a function to transition from the absolute coordinate system which is relative to the track, to the frenet space system which is relative to the vehicle
33.3	Frenet-Space Coordinates	Test implementation of function in Simulator	Waiting	Dependent on prior task	P3	Software	Localization	Simon			2025-03-08	Planning	Test the impementation of using the frenet space system in the simulator
33.4	Frenet-Space Coordinates	Test implementation of function on real car	Waiting	Dependent on prior task	P3	Software	Localization	Simon			2024-04-01	Planning	Once validated in the simulator, proceed to testing the functionnality of the Frenet-Space system on the real car and on the track